

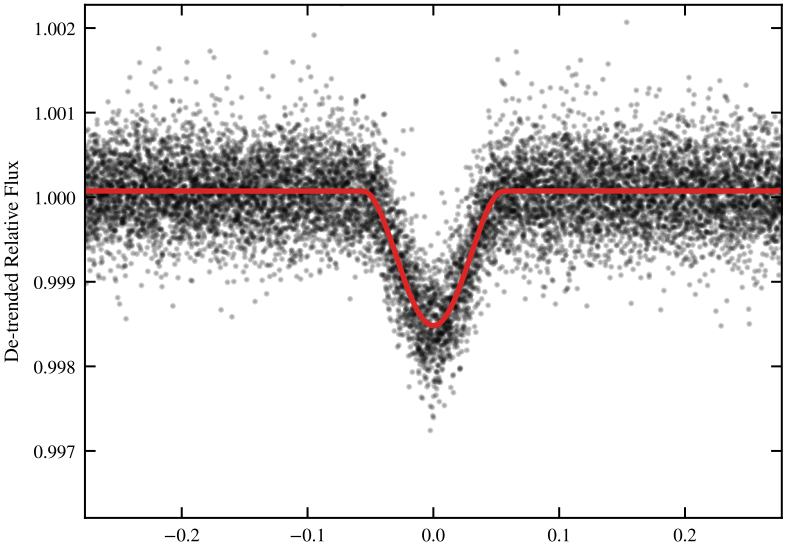
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 427508467 | RA: 117.09529° | Dec: 6.78506° | Sectors: 34

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	0.8842 ± 3.41e-05
Rp/R★	0.33933 ± 0.119
Rp (R⊕)	100.66 ± 35.6
b	1.2585 ± 0.131
T14 (h)	2.6585 ± 0.0851
a (AU)	0.021771 ± 0.00149
Teq (K)	4807.5 ± 139

Primary Phased Transit



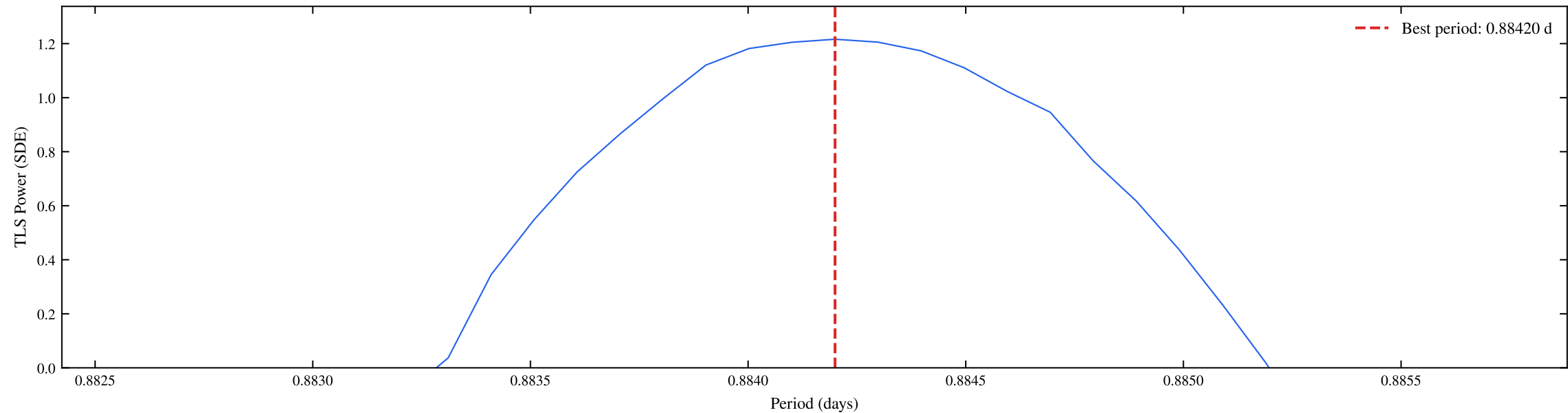
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	0.8842	0.884203 ± 0.00003
P1 Epoch (BTJD)	N/A	2229.5201 ± 0.0005
Max R-hat (R)	N/A	1.0007
Min ESS	N/A	5771
MCMC Divergences	N/A	13

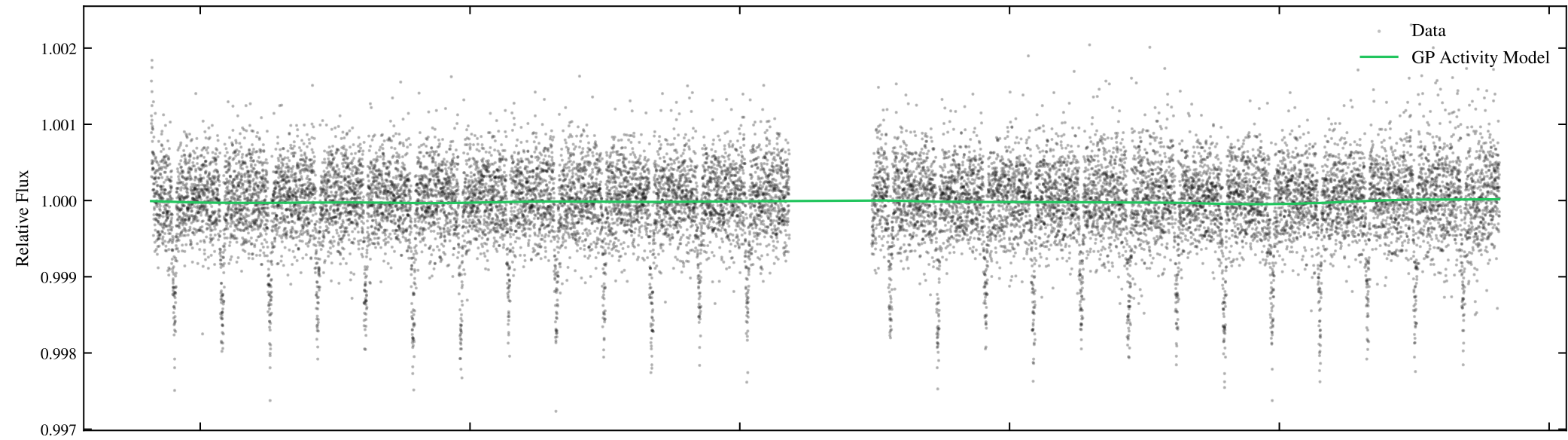
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	2.722 ± 0.12	VizieR TIC8
M★ (M☉)	2.251 ± 0.257	VizieR TIC8
Teff (K)	8928 ± 109	VizieR TIC8
log g (dex)	3.921 ± 0.0815	VizieR TIC8
[Fe/H]	N/A	Catalog
ρ★ (g/cm³)	0.1574 ± 0.0275	Derived

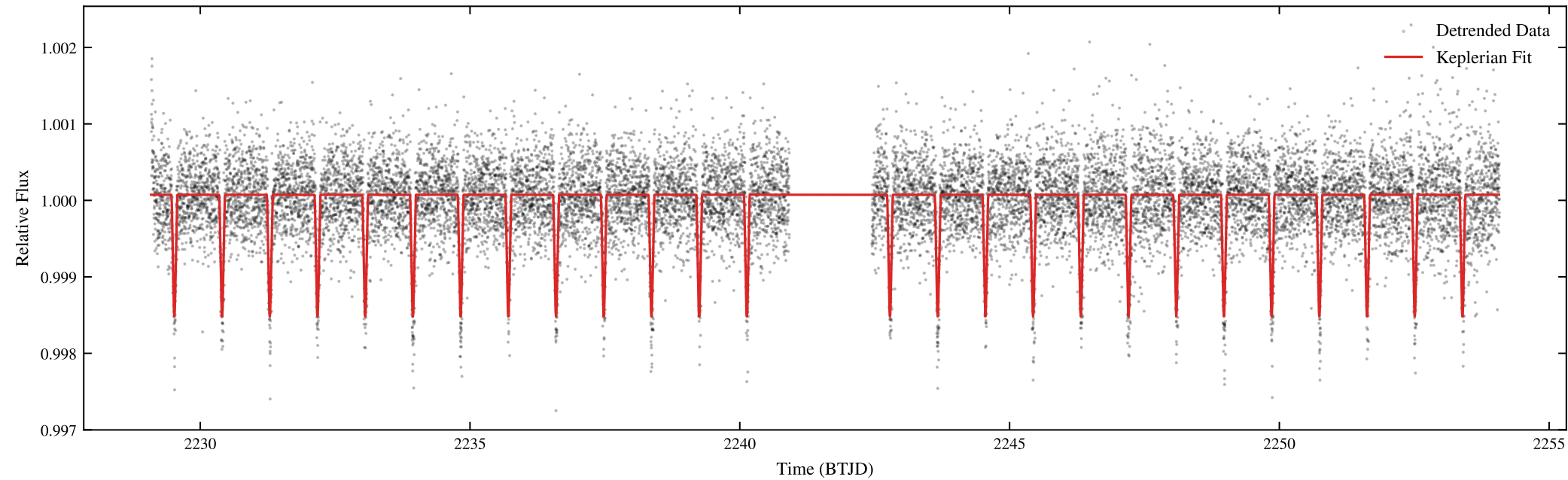
TIC 427508467 | Sectors: 34 | TLS Periodogram (SDE = 1.22)



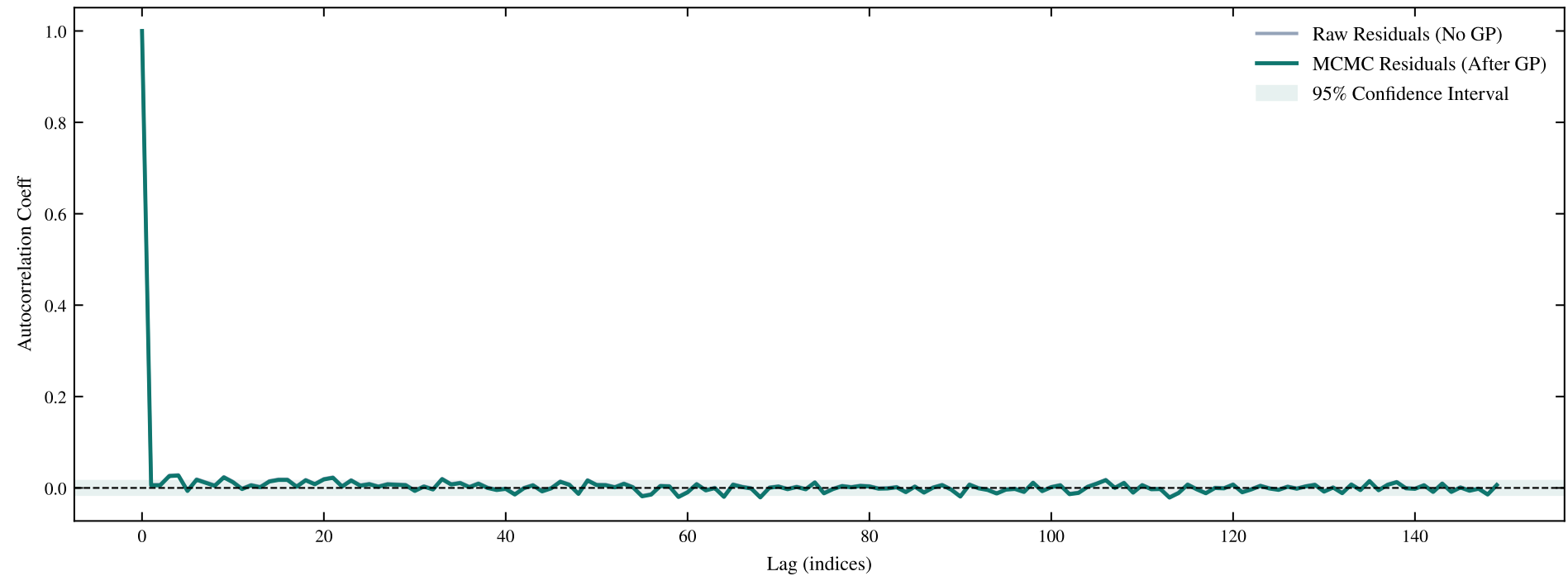
TIC 427508467 | Sectors: 34 | Bayesian GP Detrending (Time Domain)



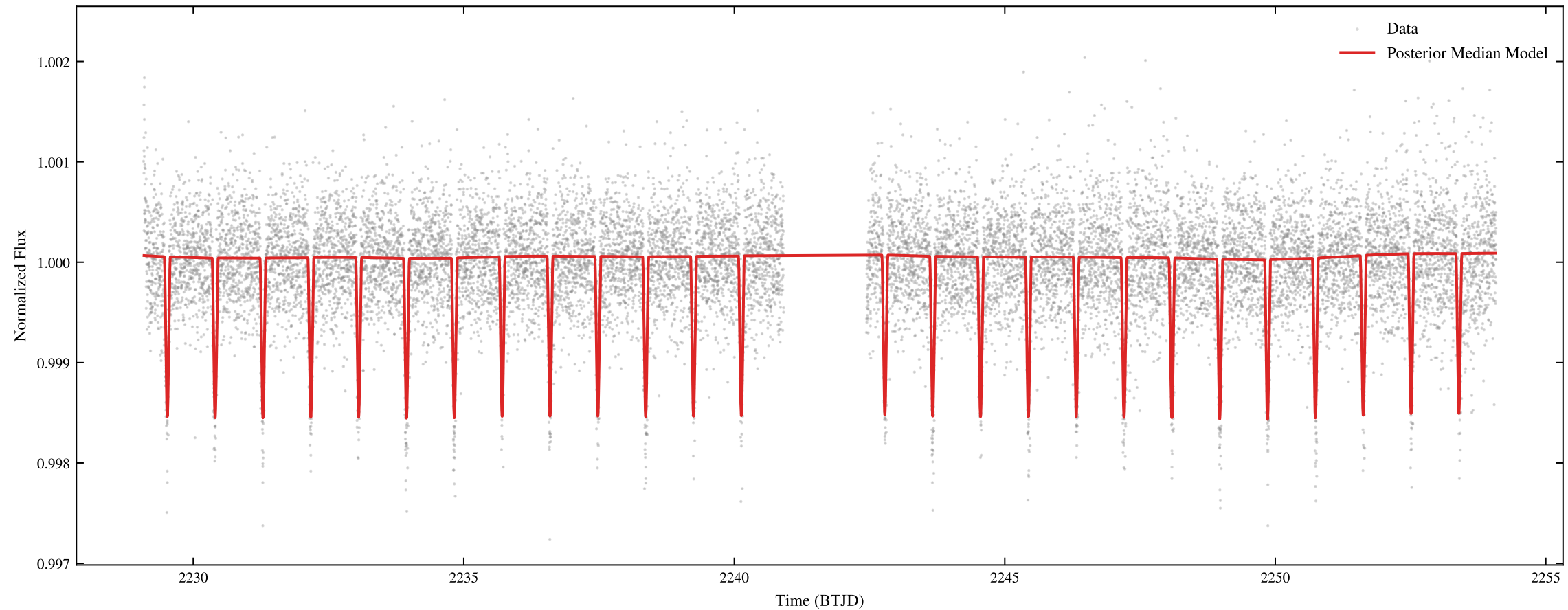
Detrended Light Curve & Best-Fit Transit Model

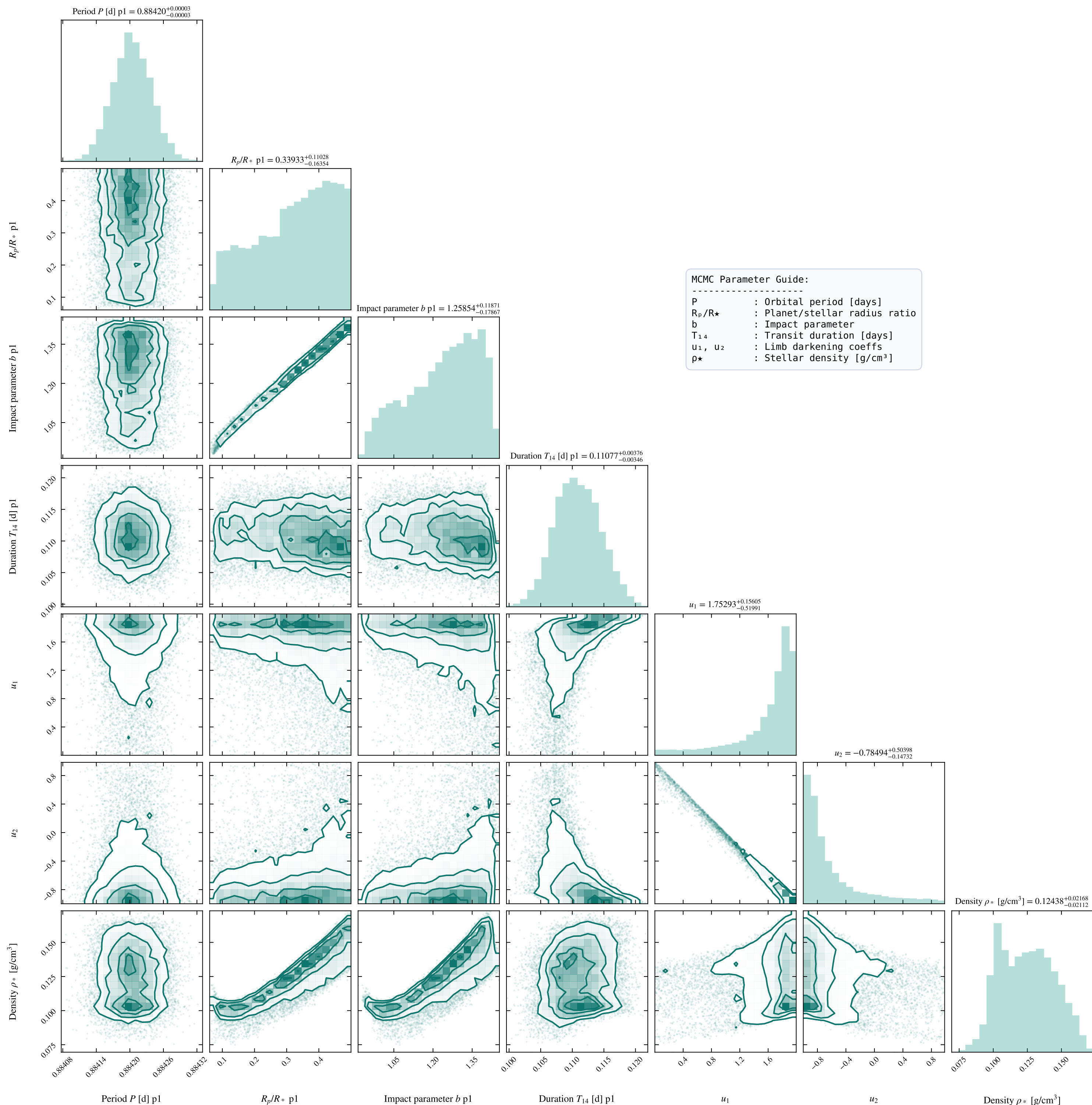


TIC 427508467 | Sectors: 34 | GP Residuals Autocorrelation Function (ACF)

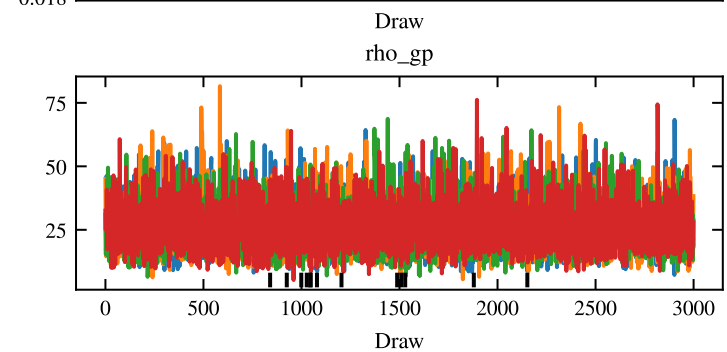
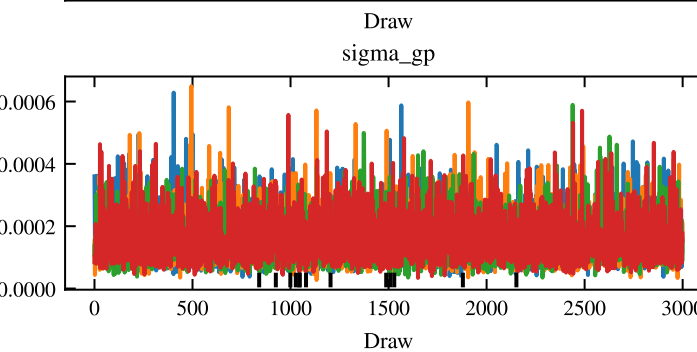
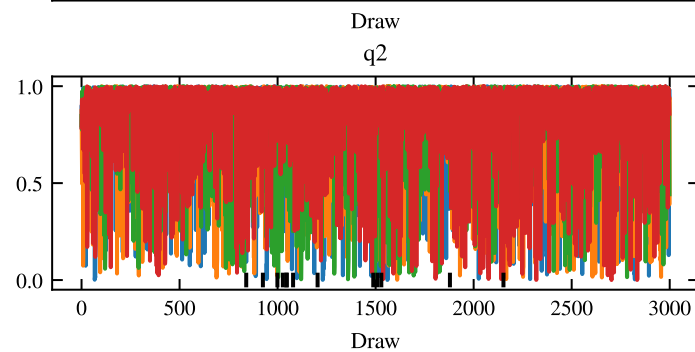
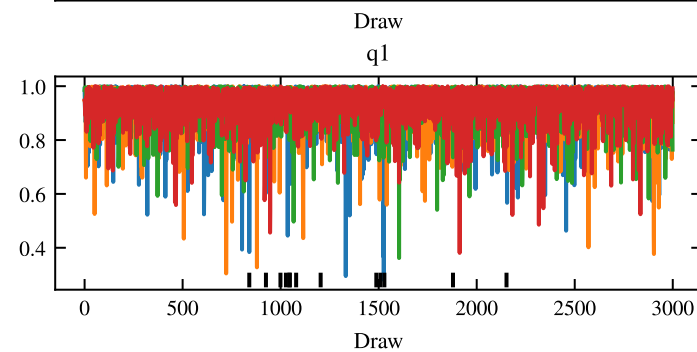
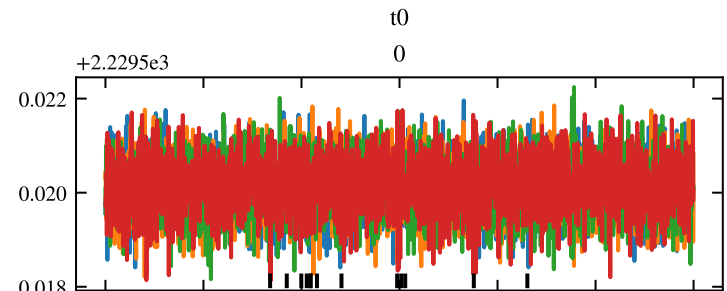
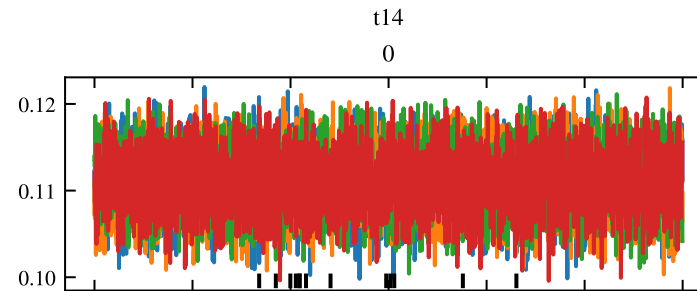
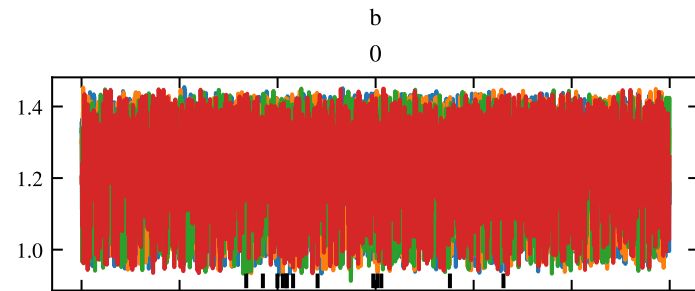
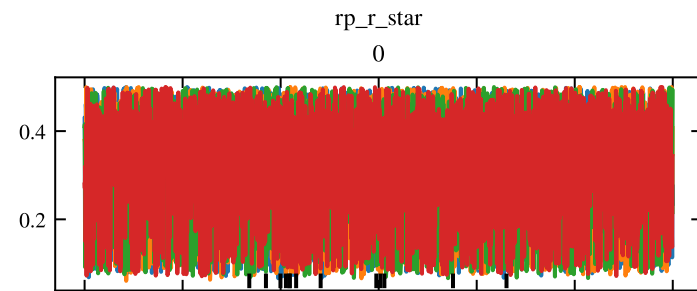


TIC 427508467 | Sectors: 34 | MCMC Posterior Predictive Check

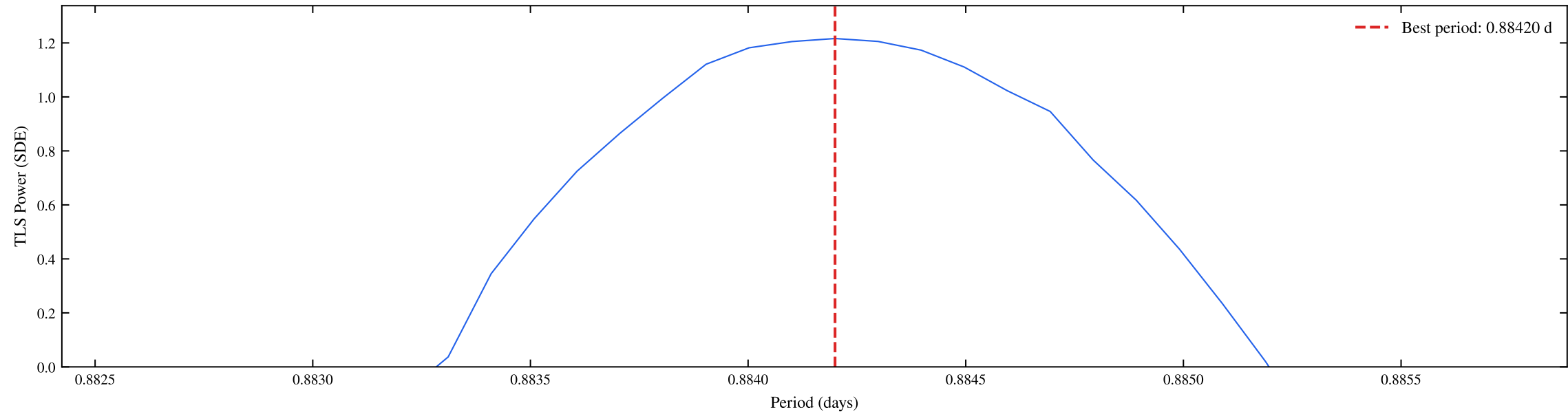




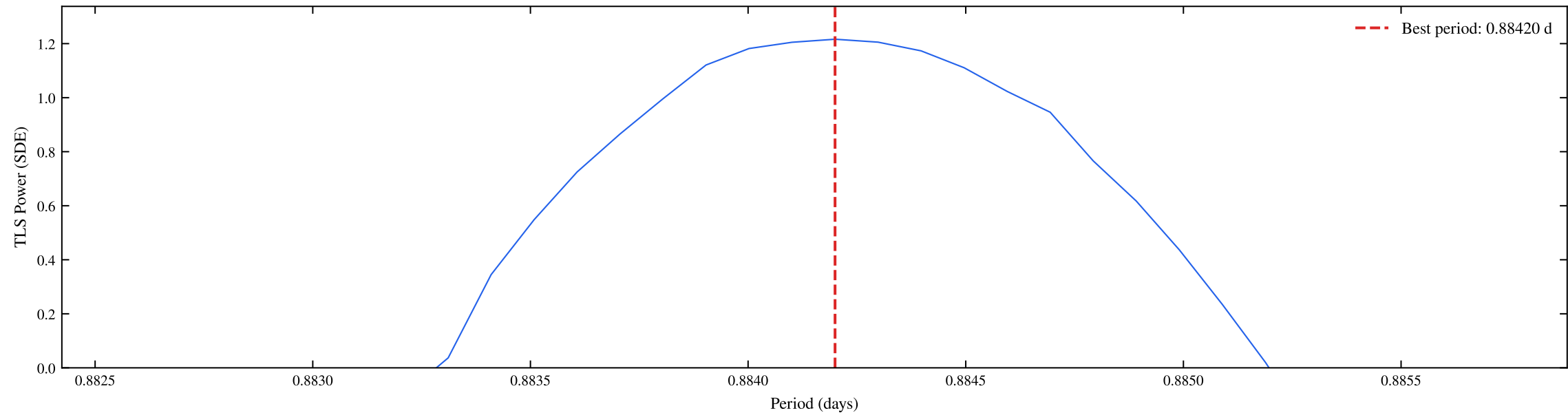
TIC 427508467 | Sectors: 34 | MCMC Trace Plots



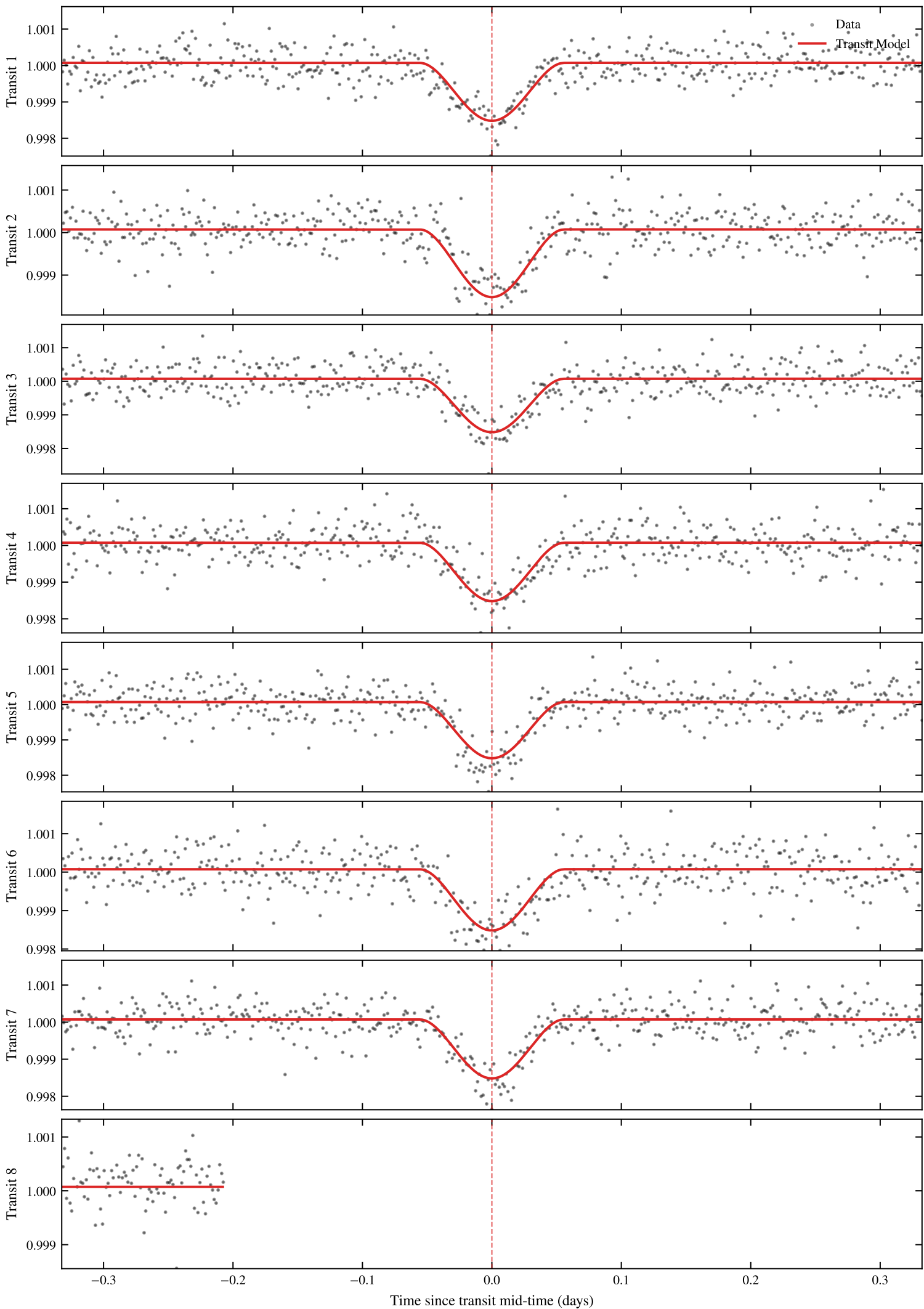
TIC 427508467 | Sectors: 34 | TLS Coarse Periodogram (SDE = 1.22)



TIC 427508467 | Sectors: 34 | TLS Fine Periodogram (SDE = 1.22)



TIC 427508467 | Sectors: 34 | Stacked Individual Transits



TIC 427508467 | Sectors: 34 | Phased MCMC Fit - Planet 1

